

HEALTH & FITNESS KNOWLEDGE BASE

First Aid • Gym Supplements • Healthcare Reference

Optimised for RAG / Vector-Store Ingestion

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PART 1 — FIRST AID ESSENTIALS

First aid is the immediate assistance given to any person suffering a sudden illness or injury, with care provided to preserve life, prevent the condition from worsening, and promote recovery. This section provides step-by-step protocols for the most common emergency scenarios.

1.1 Cardiopulmonary Resuscitation (CPR)

CPR is an emergency procedure combining chest compressions and rescue breaths to maintain circulatory flow and oxygenation during cardiac arrest. Every minute without CPR reduces survival chance by approximately 10%. Bystander CPR can double or triple survival rates.

Recognising the Need for CPR

- Person is unresponsive and does not react to tapping shoulders or calling out.
- Absent or abnormal breathing (gasping / agonal breathing counts as absent).
- No pulse detected at carotid artery within 10 seconds.

Adult CPR — Step-by-Step Protocol

- CALL: Dial emergency services (112 / 911 / 999) or direct someone specific to call.
- POSITION: Place the person on a firm, flat surface on their back.
- HANDS: Interlock hands, heel of bottom hand on centre of chest (lower half of sternum).
- COMPRESS: Push down hard and fast — at least 5–6 cm deep at 100–120 compressions/min (rate of 'Stayin' Alive').
- RECOIL: Allow full chest recoil between compressions; do not lean on the chest.
- RESCUE BREATHS (if trained): 30 compressions : 2 breaths. Tilt head, lift chin, seal lips, give 1-second breath until chest rises.
- CONTINUE: Until AED arrives, professional help takes over, person shows signs of life, or you are physically unable to continue.

Infant & Child CPR Differences

- Infants (<1 yr): Use 2 fingers on centre of chest; depth 4 cm; 30:2 ratio.
- Children (1–8 yr): One or two hands; depth one-third chest depth; 30:2 ratio.
- Cover mouth AND nose for infants; give gentler puffs.

AED (Automated External Defibrillator) Use

- Power on the AED and follow audio/visual prompts exactly.
- Attach pads as illustrated — right clavicle, left rib / apex.
- Clear the patient and allow AED to analyse rhythm; deliver shock if advised.
- Immediately resume CPR for 2 minutes after shock, then re-analyse.

Compression rate: 100–120 per minute | Depth (adult): 5–6 cm | Ratio: 30:2 | Hands-only CPR acceptable if untrained in rescue breaths | Time to first defibrillation is the strongest predictor of survival.

1.2 Choking & Airway Obstruction

Choking occurs when a foreign object partially or completely blocks the airway. Mild obstruction allows coughing; severe obstruction requires immediate intervention.

Mild Choking (Effective Cough)

- Encourage continuous coughing — do not interfere.
- Monitor closely; be ready to escalate if cough weakens.

Severe Choking — Adult (Heimlich Manoeuvre)

- Ask 'Are you choking?' — if they cannot speak, cough, or breathe, act immediately.
- Give 5 firm back blows between shoulder blades with heel of hand.
- If unsuccessful: stand behind them, fist above navel, cover with other hand, give 5 sharp inward-upward abdominal thrusts.
- Alternate 5 back blows / 5 abdominal thrusts until object expelled or person loses consciousness.
- If unconscious: lower to floor, call emergency services, begin CPR — look for object before rescue breaths.

Choking in Infants (<1 year)

- Support head; hold face-down along forearm — give 5 back blows.
- Turn face-up; 2 fingers on chest — give 5 chest thrusts (NOT abdominal thrusts).
- Look in mouth before each set; never do blind finger sweeps.

1.3 Bleeding & Wound Management

Uncontrolled bleeding is a leading preventable cause of death in trauma. Rapid, direct pressure is the cornerstone of haemorrhage control.

External Bleeding — Control Protocol

- PPE: Wear gloves if available to prevent cross-infection.
- EXPOSE: Remove or cut away clothing around the wound.
- PRESS: Apply firm, continuous direct pressure with a clean cloth or dressing — maintain for at least 10 minutes without peeking.
- ELEVATE: Raise the injured limb above heart level if fracture is not suspected.
- PACK: For deep wounds, pack tightly with gauze/clean cloth before applying pressure.
- TOURNIQUET: For life-threatening limb bleeding uncontrolled by direct pressure — apply 5–7 cm above wound, tighten until bleeding stops, note time applied.
- DRESS: Once bleeding controlled, apply pressure bandage; do not remove if blood soaks through — add more material on top.

Types of Bleeding

Type	Appearance	Risk	Management
Arterial	Bright red, spurting, pulsatile	High — rapid blood loss	Direct pressure, tourniquet if limb
Venous	Dark red, steady flow	Moderate	Direct pressure, elevation
Capillary	Oozing, slow	Low	Clean, dress, monitor
Internal	Bruising, distended abdomen, shock signs	Very high	Immobilise, treat shock, emergency services

Wound Cleaning (Minor Wounds)

- Rinse under clean running water for 5–10 minutes to remove debris.
- Gently clean around wound with mild soap; avoid soap inside wound.
- Pat dry with sterile gauze; apply antiseptic if available.
- Cover with sterile adhesive dressing; change daily or when soiled.
- Seek medical attention for: deep/gaping wounds, animal bites, puncture wounds, signs of infection (redness, pus, increasing pain, fever).

When to Call Emergency Services for Bleeding

- Blood soaking through multiple dressings and not slowing.

- Suspected internal bleeding (distended abdomen, signs of shock).
- Amputation or near-amputation injuries.
- Bleeding from head, neck, chest, or abdomen.

1.4 Burns & Scalds

Burns are tissue damage caused by heat, chemicals, electricity, friction, or radiation. Accurate assessment of depth and extent guides treatment decisions.

Classification of Burns

Degree	Depth	Appearance	Sensation	Healing
Superficial (1st)	Epidermis only	Red, dry, no blisters	Painful	3–5 days, no scarring
Partial thickness (2nd)	Epidermis + dermis	Red, moist, blisters	Very painful	2–3 weeks, possible scarring
Full thickness (3rd)	All skin layers	White/brown/black, leathery	Painless (nerve destroyed)	Requires grafting
4th degree	Muscle/bone involved	Charred, blackened	Painless	Requires surgery/amputation

Immediate First Aid for Burns

- COOL: Run cool (not ice-cold) water over burn for minimum 20 minutes — start within 3 hours of injury.
- COVER: Use a clean, non-fluffy, non-stick material (cling film is ideal) loosely over the burn.
- Do NOT: burst blisters, apply butter/toothpaste/ice/creams.
- Do NOT: remove clothing stuck to burn.
- ANALGESIA: Give appropriate pain relief if available.
- HYDRATE: Encourage oral fluids for minor burns; IV fluids needed for major burns.

Rule of Nines — Body Surface Area Estimation

Head & neck = 9% | Each arm = 9% | Chest = 9% | Abdomen = 9% | Upper back = 9% | Lower back = 9% | Each thigh = 9% | Each lower leg = 9% | Genitalia = 1%

- Seek hospital care for: burns >10% BSA in adults (>5% in children), face/hands/feet/genitalia burns, circumferential burns, electrical/chemical burns, any 3rd degree burn.

1.5 Fractures & Sprains

A fracture is a break in bone continuity. A sprain is ligament stretching or tearing. Both can present similarly — when in doubt, treat as a fracture.

Recognising a Fracture (DOTS Mnemonic)

- D — Deformity: abnormal shape or position compared to opposite limb.
- O — Open wounds: skin broken near suspected fracture site.
- T — Tenderness: localised pain on palpation along bone.
- S — Swelling/bruising: appears rapidly over injury site.

RICE Protocol (Sprains & Soft Tissue)

- R — Rest: stop activity immediately; avoid weight-bearing.
- I — Ice: apply ice pack (wrapped in cloth) for 20 min every 2 hours for first 48 hrs.
- C — Compression: apply elastic bandage to limit swelling; not so tight it impairs circulation.
- E — Elevation: raise limb above heart level when resting.

Fracture Immobilisation

- Immobilise the joint above AND below the fracture.
 - Use improvised splints (boards, rolled magazines) padded with soft material.
 - Check circulation, sensation, movement (CSM) before and after splinting.
 - Open fractures: cover wound with sterile dressing; do NOT push bone back in.
 - Suspected spinal injury: do not move; maintain inline stabilisation until emergency services arrive.
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1.6 Shock & Anaphylaxis

Shock

Shock is a life-threatening condition where insufficient blood flow fails to meet the body's needs. Types: hypovolaemic (blood loss), cardiogenic (heart failure), distributive (sepsis/anaphylaxis), obstructive.

- Signs: pale/cold/clammy skin, rapid weak pulse, rapid breathing, confusion, thirst, nausea.
- Lay person flat; raise legs 30 cm if no spinal/chest injury.
- Keep warm with blanket; do not give food or drink.
- Treat underlying cause; monitor vital signs; call emergency services.

Anaphylaxis — Severe Allergic Reaction

- Signs: sudden rash/urticaria, swelling (face/lips/throat), difficulty breathing, wheeze, rapid pulse, drop in blood pressure, collapse.
 - ADRENALINE (EPINEPHRINE): Use auto-injector (e.g., EpiPen) into outer mid-thigh — can be given through clothing.
 - POSITION: Sit up if breathing difficulty; lie flat with legs raised if shock; recovery position if unconscious.
 - CALL 999/112: Anaphylaxis is always a medical emergency — second dose after 5–15 min if no improvement.
 - OBSERVE for biphasic reaction (symptoms returning hours later — hospital admission required).
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1.7 Head Injuries & Concussion

Head injuries range from minor scalp lacerations to severe traumatic brain injuries. Any loss of consciousness, confusion, or vomiting following head trauma requires medical evaluation.

- CONCUSSION signs: brief loss of consciousness, confusion, headache, dizziness, nausea, memory gap, light/noise sensitivity.
- REST: No return to sport/activity same day — follow graduated return-to-play protocol.
- MONITOR for 24 hours: wake every 2 hours to assess; seek emergency care if worsening headache, repeated vomiting, seizures, unequal pupils, weakness, unusual drowsiness.
- Scalp lacerations: apply firm pressure; scalp bleeds profusely due to rich blood supply — this can be alarming but not always indicative of serious injury.

Red Flag Head Injury Signs (Call 999 Immediately)

- Increasing headache not relieved by paracetamol.
- Repeated vomiting (>3 times).
- Seizure after head injury.
- Unequal, dilated or unreactive pupils.
- Clear fluid from nose or ears (CSF leak).
- Loss of consciousness >1 minute.

1.8 Heat Stroke & Hypothermia

Heat Stroke (Emergency)

- Core temperature $>40^{\circ}\text{C}$; hot/dry or flushed skin; confusion; rapid strong pulse; may stop sweating.
- COOL IMMEDIATELY: Move to shade, remove clothing, apply cool water to skin, fan vigorously, ice packs to neck/armpits/groin.
- Target: reduce core temp to $<39^{\circ}\text{C}$ within 30 minutes.
- Do NOT give fluids if unconscious; call emergency services.

Heat Exhaustion

- Heavy sweating, pale/cool/clammy skin, nausea, weakness, dizziness, headache — core temp normal or mildly elevated.
- Move to cool area, lie down, loosen clothing, cool compresses, oral rehydration fluids with electrolytes.

Hypothermia

- Core temperature $<35^{\circ}\text{C}$; shivering (absent in severe), confusion, slurred speech, stumbling, drowsiness.
 - Remove wet clothing; insulate with dry blankets; give warm sweet drinks if conscious.
 - Do NOT rub limbs or apply direct heat; handle gently (risk of cardiac arrhythmia).
 - Severe hypothermia — horizontal evacuation; CPR if no pulse; hospital for rewarming.
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1.9 Fainting & Seizures

Fainting (Syncope)

- Brief loss of consciousness due to reduced cerebral blood flow.
- Lay person flat, raise legs 30 cm; loosen tight clothing.
- Should recover within 1 minute; investigate causes (dehydration, overheating, cardiac).

Seizures (Convulsions)

- PROTECT: Clear surrounding area of hazards; do NOT restrain the person.
 - CUSHION: Protect head with folded jacket/hands.
 - TIME: Note duration; most tonic-clonic seizures stop within 1–3 minutes.
 - RECOVERY: Once jerking stops, place in recovery position; stay and reassure.
 - CALL 999 IF: first-ever seizure, duration >5 minutes, no recovery within 5 minutes, repeated seizures, injury sustained, pregnant, or diabetic.
 - NEVER: put anything in the mouth or between teeth.
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1.10 Eye & Ear Emergencies

Eye Injuries

- Chemical splash: irrigate immediately with large volumes of clean water for 15–20 minutes; seek emergency care.
- Foreign body (superficial): flush with clean water; do not rub eye; seek care if not resolved.
- Penetrating eye injury: DO NOT remove object; cover both eyes loosely; immediate hospital.

Ear Emergencies

- Insect in ear: tilt ear up, pour lukewarm (not hot) water — do not probe.
- Foreign object: do not insert anything; seek medical removal.
- Bleeding from ear after head trauma: cover loosely — possible temporal bone fracture — seek emergency care.

PART 2 — GYM SUPPLEMENTS

This section provides evidence-based information on popular gym and fitness supplements, including mechanisms of action, proven benefits, recommended dosing, and side effects. All information is for educational purposes; consult a healthcare professional before starting any supplement regimen.

2.1 Creatine Monohydrate

Creatine is one of the most extensively researched sports supplements. It is a naturally occurring compound synthesised from amino acids arginine, glycine, and methionine, primarily stored in skeletal muscle as phosphocreatine. It is also obtained from red meat and seafood.

Mechanism of Action

Creatine increases the phosphocreatine pool in muscles, rapidly regenerating ATP (adenosine triphosphate) during high-intensity, short-duration exercise (0–30 seconds). This delays fatigue and enhances repeated bout performance.

Evidence-Based Benefits

- Increases maximal strength and power output in resistance-trained individuals.
- Improves performance in repeated sprint and high-intensity interval efforts.
- Promotes lean muscle mass gains, partly via intramuscular water retention.
- Enhances muscular endurance capacity (>15% improvement reported in meta-analyses).
- Emerging evidence for cognitive benefits — may improve short-term memory and reasoning.
- Potential neuroprotective and bone-health benefits under investigation.

Dosing Protocols

Protocol	Loading Phase	Maintenance	Notes
Standard Loading	20 g/day (4 x 5 g) for 5–7 days	3–5 g/day	Faster saturation (~7 days)
No-Load Protocol	None	3–5 g/day	Full saturation in ~28 days; fewer GI issues
Cycling (not required)	Optional	3–5 g/day for 8–12 weeks, 4-week break	No evidence washout improves efficacy

Side Effects

- Weight gain of 1–2 kg in first week (water retention in muscle — not fat).
- Gastrointestinal discomfort if large doses taken at once — split doses or use no-load protocol.
- Mild cramping occasionally reported; ensure adequate hydration.
- Elevated serum creatinine (not creatine — a related metabolite) may give false kidney function concern in blood tests; communicate supplement use to doctor.

Creatine Safety Profile

Creatine monohydrate is considered safe for healthy adults. Evidence of kidney damage in healthy individuals is not supported by current research. Not recommended for individuals with pre-existing kidney disease without physician guidance. Safe for long-term use (years). Creatine HCl and buffered forms offer no proven advantage over monohydrate.

2.2 Whey Protein

Whey protein is a complete protein derived from milk during cheese production. It contains all nine essential amino acids and is particularly rich in leucine, the primary trigger for muscle protein synthesis (MPS). It is the most popular protein supplement worldwide.

Types of Whey Protein

Type	Protein Content	Lactose	Best For
Whey Concentrate (WPC)	70–80%	Moderate	General use, budget-friendly
Whey Isolate (WPI)	90–95%	Very low	Lactose-sensitive, leaning phase
Whey Hydrolysate (WPH)	>90%	Very low	Fastest absorption, post-surgery

Evidence-Based Benefits

- Stimulates muscle protein synthesis (MPS) more effectively than other protein sources due to high leucine content.
- Supports muscle recovery and reduces delayed onset muscle soreness (DOMS) when consumed post-exercise.
- Promotes satiety and may support fat loss when replacing higher-calorie foods.
- Convenient method to meet daily protein targets (1.6–2.2 g/kg body weight for muscle building).
- Immunoglobulins and lactoferrin in whey may support immune function.

Dosing Guidelines

- Optimal single dose: 20–40 g per serving (0.4 g/kg body weight) to maximise MPS.
- Post-workout window: consume within 0–2 hours post-training (total daily protein matters more than timing).
- Distribute protein intake across 3–5 meals/snacks (~25–40 g each) for optimal daily MPS.

Side Effects

- Bloating and gas — often due to lactose (switch to isolate) or additives.
- Acne in susceptible individuals — whey may increase IGF-1 and insulin levels.
- Allergic reactions in those with cow's milk protein allergy — use plant protein instead.
- Excessive use replacing whole foods can lead to nutritional imbalances (fibre, micronutrients).
- High doses (>3 g/kg/day consistently) may stress kidneys in pre-existing kidney disease.

2.3 Multivitamins

A multivitamin is a supplement containing a combination of vitamins and minerals. While they cannot replace a varied whole-food diet, they can help fill nutritional gaps particularly in athletes with high energy demands, restricted diets, or poor dietary variety.

Key Micronutrients in Athletic Performance

Nutrient	Role in Body	Deficiency Risk	RDA (Adult)
Vitamin D3	Bone health, immunity, testosterone support	Very common (indoor lifestyle)	600–2000 IU/day
Vitamin C	Collagen synthesis, antioxidant, immunity	Common in athletes	75–90 mg/day
B12	Red blood cell production, nerve function	Vegetarians/vegans	2.4 mcg/day
Iron	Oxygen transport (haemoglobin)	Female athletes, distance runners	8–18 mg/day
Magnesium	Muscle function, sleep, 300+ enzymes	Common in hard trainers	310–420 mg/day
Zinc	Immune function, testosterone, protein synthesis	Common with heavy sweating	8–11 mg/day
Folate (B9)	DNA synthesis, cell division	Pregnant women	400 mcg/day
Calcium	Bone density, muscle contraction	Dairy-free diets	1000–1200 mg/day

Benefits of Multivitamin Supplementation

- Corrects micronutrient deficiencies that impair energy metabolism, immunity, and recovery.

- May reduce risk of upper respiratory infections in intensely training athletes.
- Supports bone health in high-impact sport athletes.
- Convenient baseline nutritional insurance for individuals with poor dietary diversity.

Side Effects & Cautions

- Fat-soluble vitamins (A, D, E, K) can accumulate — toxicity risk with mega-doses.
- Vitamin A excess: headaches, liver damage, birth defects (teratogenic).
- Iron excess: constipation, oxidative stress; men should avoid high-iron formulas unless deficient.
- Some compounds compete for absorption (calcium & iron, zinc & copper) — timing and balance matter.
- Nausea if taken on empty stomach — take with food.

Multivitamin Best Practices

- Choose a product third-party tested (NSF, Informed Sport, USP certified).
- Look for methylated B vitamins (methylfolate, methylcobalamin) for better bioavailability.
- Avoid mega-dose formulations unless specifically recommended by a clinician.
- Whole-food vitamins (from fermented foods) may offer better absorption than synthetic.

2.4 Pre-Workout Supplements (Caffeine-Based)

Pre-workout formulas typically combine caffeine with other ergogenic compounds. Caffeine is the most evidence-supported performance-enhancing substance available.

Common Ingredients & Evidence

Ingredient	Dose	Benefit	Evidence Level
Caffeine	3–6 mg/kg BW	Focus, power, endurance, fat oxidation	A (very strong)
Beta-Alanine	3.2–6.4 g/day	Muscular endurance (>60s bouts)	A (strong)
L-Citrulline	6–8 g	Blood flow, endurance, reduced soreness	B (moderate-strong)
Creatine	3–5 g	Power, strength (see 2.1)	A (very strong)
B-Vitamins	Varied	Energy metabolism	B (modest)
Taurine	1–2 g	Reduces oxidative stress, cramping	B (moderate)
Tyrosine	500–2000 mg	Focus under stress/sleep deprivation	B (moderate)

Benefits

- Caffeine reduces perception of effort (RPE) and enhances power output by 3–7%.
- Improves athletic endurance — particularly aerobic performance.
- Enhances mental focus, reaction time, and alertness.
- May increase fat oxidation during exercise.

Side Effects

- Caffeine: insomnia (avoid within 6 hours of sleep), anxiety, elevated heart rate, GI distress, dependence/tolerance.
- Beta-alanine: harmless but uncomfortable paraesthesia (tingling/flushing of skin) — reduced by splitting doses.
- High stimulant combinations (multiple caffeine sources) risk palpitations, hypertension.
- Proprietary blends often underdose key ingredients — check label for full disclosed amounts.

Pre-Workout Safety

- Not suitable for individuals with heart conditions, anxiety disorders, or hypertension.
- Avoid combining with other caffeine sources (coffee, energy drinks, fat burners).
- Cycle use to prevent tolerance — 8 weeks on, 2 weeks off.
- Check for banned substances if competing under WADA/USADA — use Informed Sport certified products.

2.5 BCAAs — Branched-Chain Amino Acids

BCAAs are three essential amino acids: Leucine, Isoleucine, and Valine. They account for approximately 35% of the essential amino acids in muscle protein and are metabolised directly in muscle (unlike other AAs processed in the liver).

- Leucine: most anabolic BCAA; directly activates mTORC1 signalling pathway for MPS.
- Isoleucine: promotes glucose uptake into muscle cells; may support immune function.
- Valine: reduces central nervous system fatigue by competing with tryptophan for brain entry.

Benefits

- Reduces muscle protein breakdown (catabolism) during fasted training.
- May reduce DOMS and muscle damage markers (CK, LDH) post-exercise.
- Useful for prolonged training bouts (>90 minutes) or calorie-restricted phases.

Limitations & Side Effects

- If total daily protein is adequate (>1.6 g/kg), additional BCAA supplementation offers minimal benefit.
- Generally very safe; excess excretion via urine if above requirements.
- Rare GI discomfort; some sweeteners in BCAA products cause digestive issues.

BCAA vs Whey Protein

Whey protein contains all essential amino acids INCLUDING BCAAs (~25% by weight). For those already meeting protein targets, standalone BCAAs are largely redundant. BCAAs have value when training fasted or when whole protein sources are unavailable. Ideal BCAA ratio: 2:1:1 (Leucine:Isoleucine:Valine).

2.6 Beta-Alanine

Beta-alanine is a non-essential amino acid that combines with histidine to form carnosine in muscle tissue. Carnosine acts as an intracellular pH buffer, delaying the acidosis that contributes to muscle fatigue during high-intensity exercise lasting 60–240 seconds.

- Benefits: Increased work capacity in the 1–4 minute effort range; improved sprint performance; reduced fatigue in CrossFit-style training.
- Effective for: rowing, cycling, swimming, team sports with repeated sprints, metabolic conditioning.
- Less effective for: pure strength/powerlifting (<10 seconds), pure endurance (>10 min).
- Side effects: Paraesthesia (skin tingling/flushing) — harmless, dose-dependent; minimised by splitting doses to <1.6 g per serving or using sustained-release forms.
- Dosing: 3.2–6.4 g/day in split doses; 4–6 weeks to achieve significant carnosine elevation.

2.7 Fish Oil / Omega-3 Fatty Acids

Fish oil provides EPA (eicosapentaenoic acid) and DHA (docosahexaenoic acid), long-chain omega-3 polyunsaturated fatty acids with extensive health and performance research.

Benefits

- Cardiovascular health: reduces triglycerides by 15–30%; supports healthy blood pressure and endothelial function.
- Anti-inflammatory: EPA competes with arachidonic acid; reduces inflammatory cytokines.
- Muscle recovery: 3–4 g/day EPA+DHA may reduce DOMS and muscle protein breakdown.
- Brain & mood: DHA is structural component of brain (60% of brain lipids); supports cognitive function, may reduce depression.
- Joint health: anti-inflammatory benefits support joint comfort in high-volume training.

Dosing & Side Effects

- General health: 1–2 g EPA+DHA/day. Performance/anti-inflammatory: 3–4 g/day.
 - Take with meals to minimise fishy burps; enteric-coated capsules reduce GI discomfort.
 - High doses (>3 g/day) may slightly increase bleeding time — caution with blood thinners.
 - Vegans: algae-based DHA/EPA is equally effective (fish get omega-3 from algae).
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2.8 Vitamin D3 & K2

Vitamin D3 (cholecalciferol) is a fat-soluble vitamin produced in skin upon UVB exposure. Estimated 1 billion people globally are deficient. K2 (menaquinone) works synergistically to direct calcium to bones rather than arteries.

- Benefits of D3: Bone mineralisation; testosterone synthesis support; immune modulation; mood regulation; muscle function.
- Benefits of K2: Activates osteocalcin (directs calcium into bone); activates matrix Gla protein (removes calcium from arteries).
- Athletes training indoors or in low-sun climates are at particular risk of deficiency.

Dosing & Safety

- Maintenance: 1000–2000 IU D3 daily; correction of deficiency: 4000–6000 IU/day under medical supervision.
 - K2 (MK-7 form): 100–200 mcg/day alongside D3 supplementation.
 - Toxicity: rare but possible at very high chronic doses (>10,000 IU/day D3) — hypercalcaemia, kidney stones.
 - K2 contraindicated with warfarin anticoagulants — consult doctor.
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2.9 Magnesium

Magnesium is the fourth most abundant mineral in the body, cofactor in over 300 enzymatic reactions. Deficiency is widespread — estimated in 50–60% of the general population, and more prevalent in athletes who lose magnesium through sweat.

- Benefits: Muscle relaxation and cramp prevention; ATP production; sleep quality improvement; protein synthesis; stress/cortisol regulation; blood glucose control.
 - Types: Magnesium glycinate (best for sleep/anxiety — high absorption, gentle); citrate (moderate absorption, laxative effect); oxide (poor absorption); malate (energy, muscle recovery); threonate (crosses blood-brain barrier).
 - Dosing: 200–400 mg elemental magnesium daily; best taken in evening with food.
 - Side effects: Loose stools/diarrhoea at high doses (especially oxide/citrate) — reduce dose or switch form.
 - Caution in kidney impairment — cannot excrete excess magnesium effectively.
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2.10 Ashwagandha (Adaptogen)

Ashwagandha (*Withania somnifera*) is an Ayurvedic adaptogenic herb. Adaptogens help the body resist physical, chemical, and biological stressors. Key active compounds are withanolides.

- Benefits: Reduces cortisol by 14–30% in clinical trials; improves stress/anxiety scores; enhances recovery from training; may increase testosterone in men with low-normal levels; improves VO2 max; improves sleep quality.
 - Dosing: 300–600 mg/day of root extract (KSM-66 or Sensoril standardised extract); can be taken anytime with food.
 - Side effects: Generally well tolerated; rare GI upset; sedation at high doses; rare liver injury cases reported with very high doses.
 - Contraindications: Autoimmune conditions (may stimulate immune system); thyroid disorders; pregnant women.
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2.11 L-Glutamine

L-Glutamine is the most abundant amino acid in blood and muscle tissue. It is conditionally essential — demand exceeds synthesis capacity during periods of intense physiological stress.

- Benefits: Primary fuel source for intestinal epithelial cells and immune cells; supports gut barrier integrity; may reduce GI permeability ('leaky gut') under heavy training stress; supports immune function in overtrained athletes; aids recovery from surgery or critical illness.
 - Athletic benefit: Evidence is mixed for performance gains in healthy athletes meeting protein needs; more beneficial for overtrained individuals or those on very low calorie diets.
 - Dosing: 5–10 g/day; post-workout or before bed.
 - Side effects: Very safe; no known serious adverse effects at recommended doses; excess converted to glucose or excreted.
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2.12 Zinc & ZMA

Zinc is an essential trace mineral involved in over 300 enzyme reactions, immune function, protein synthesis, and testosterone production. ZMA is a combination of zinc, magnesium aspartate, and vitamin B6, marketed primarily for recovery and testosterone support.

- Benefits: Immune function; wound healing; testosterone production (deficiency blunts T levels); protein synthesis; antioxidant enzyme function (superoxide dismutase).
- Athletes who sweat heavily or follow plant-based diets are most at risk of deficiency.
- ZMA evidence: Improves sleep quality and recovery; testosterone benefit is likely limited to correcting deficiency, not supranormal supplementation.
- Dosing: Zinc 8–25 mg/day (RDA 8–11 mg); ZMA per label instructions (usually taken pre-sleep on empty stomach).
- Side effects: Nausea on empty stomach; high zinc (>40 mg/day long-term) impairs copper absorption — ensure zinc:copper ratio ~10:1; metallic taste at high doses.

PART 3 — GENERAL HEALTHCARE REFERENCE

3.1 Hydration & Electrolytes

Water accounts for 60% of adult body weight and is essential for thermoregulation, nutrient transport, joint lubrication, and cognitive function. Even 2% dehydration impairs athletic performance and cognitive function.

- Baseline: ~35 ml/kg body weight/day; ~500 ml extra per hour of moderate exercise.
- Urine colour guide: Pale yellow = well hydrated; dark yellow/amber = dehydrated; clear = overhydrated (dilutes electrolytes).
- Electrolytes lost in sweat: Sodium (largest), Chloride, Potassium, Magnesium, Calcium.
- Sodium: 1000–2000 mg/hour during intense prolonged exercise; prevents hyponatraemia in endurance events.
- Hyponatraemia (low sodium): more dangerous than dehydration; caused by drinking too much plain water during long events; symptoms: nausea, headache, confusion, seizures.
- Oral Rehydration Solution (ORS): 1 litre water + 6 tsp sugar + 0.5 tsp salt — WHO formula for rapid rehydration.

3.2 Sleep Hygiene & Recovery

Sleep is the most powerful recovery tool available. During deep (slow-wave) sleep, growth hormone peaks, muscle repair occurs, glycogen is restored, and memories consolidate. Chronic sleep deprivation is associated with reduced testosterone, elevated cortisol, increased injury risk, and impaired body composition.

- Athletes require 8–10 hours per night; general adult recommendation is 7–9 hours.
- Sleep architecture: 4–6 cycles per night; each ~90 minutes; early cycles are deep sleep dominant; later cycles REM-dominant.

Sleep Hygiene Best Practices

- Consistent sleep/wake time (even weekends) — anchors circadian rhythm.
- Cool room temperature: 17–19°C (62–67°F) optimal for sleep onset.
- Eliminate blue light 60–90 min before bed (phones, screens) or use blue-light-blocking glasses.
- Avoid caffeine within 6 hours of sleep; avoid alcohol within 3 hours (disrupts REM).
- Dark, quiet environment; blackout curtains; white noise if needed.
- Pre-sleep nutrition: 40 g casein protein may support overnight MPS; tart cherry juice may improve sleep quality.
- Supplements for sleep: Magnesium glycinate (200–400 mg), Ashwagandha (300 mg), Melatonin (0.5–3 mg for jet lag/shift work — not chronic insomnia treatment).

3.3 Inflammation & Recovery

Acute inflammation following exercise is a necessary and beneficial signal for muscle adaptation. Excessive or chronic inflammation, however, impairs recovery and health.

- Beneficial post-exercise inflammation: peaks at 24–48 hrs; initiates satellite cell activation, protein synthesis signals.
- Do NOT suppress all exercise-induced inflammation: routine high-dose NSAIDs (ibuprofen) or antioxidant supplements post-workout may blunt adaptation.

Anti-Inflammatory Strategies

- Dietary: Increase omega-3 fatty acids, turmeric/curcumin, tart cherry, ginger, dark berries, olive oil.
- Sleep: Inadequate sleep markedly elevates inflammatory markers (IL-6, CRP, TNF-alpha).
- Stress management: Chronic cortisol elevation promotes systemic inflammation.

- Exercise programming: Deloads, periodisation, and rest days are essential for managing cumulative inflammatory load.
- Recovery modalities: Cold water immersion (reduces acute inflammation, may blunt hypertrophy); sauna (reduces chronic inflammation markers); massage; foam rolling.

3.4 Gut Health & Probiotics

The gut microbiome — 38 trillion bacteria in the GI tract — profoundly affects digestion, immunity, mental health, inflammation, and even athletic performance. A diverse, healthy microbiome is associated with better health outcomes.

- Probiotics: Live microorganisms providing health benefits when consumed in adequate amounts. Common strains: Lactobacillus acidophilus, L. rhamnosus GG, Bifidobacterium longum.
- Prebiotic fibre: Feeds beneficial bacteria — sources: garlic, onions, leeks, asparagus, bananas, oats, legumes.
- Benefits of probiotics: Reduced GI upset in athletes; shorter/milder URIs; improved lactose digestion; potential mood benefits via gut-brain axis.
- Dosing: ≥10 billion CFU/day; multi-strain formulas preferred; take with food; refrigerate live cultures.
- Fermented foods (kefir, yoghurt, kimchi, sauerkraut, kombucha) provide natural probiotic strains.
- Side effects: Initial bloating/gas normalises within 1–2 weeks; not suitable without medical supervision for severely immunocompromised individuals.

3.5 Mental Wellness & Stress Management

Mental health is inseparable from physical performance and recovery. Chronic psychological stress elevates cortisol, impairs sleep, suppresses immunity, and increases injury risk.

- HPA Axis: Hypothalamic-Pituitary-Adrenal axis governs stress response. Chronic activation leads to adrenal fatigue, hormonal imbalances, and systemic inflammation.
- Exercise: 150 min/week moderate aerobic exercise reduces depression risk by 30–40%; releases BDNF (brain-derived neurotrophic factor) — 'Miracle-Gro for the brain'.
- Mindfulness/meditation: Even 10 minutes daily reduces cortisol; improves focus and emotional regulation.
- Social connection: Loneliness is as damaging to health as smoking 15 cigarettes/day (Holt-Lunstad, 2015).
- Supplements with evidence for stress: Ashwagandha (see 2.10), Magnesium (see 2.9), L-theanine (100–200 mg — promotes calm focus), Rhodiola rosea (150–400 mg — adaptogen for mental fatigue).

3.6 Common Supplement Interactions

Combination	Interaction	Recommendation
Calcium + Iron	Compete for same transporter — calcium reduces iron absorption by up to 60%	Separate by 2 hours
Zinc + Copper	High zinc depletes copper stores (same transporter)	Maintain 10:1 Zn:Cu ratio or take separately
Vitamin D3 + K2	Synergistic — D3 increases calcium absorption	Take together with fat-containing meal
Caffeine + Creatine	No significant negative interaction; may slightly increase creatine loading	Generally safe to combine
Omega-3 + Blood Thinners	Omega-3 has mild antiplatelet effects — additive with aspirin	Use with caution
Iron + Vitamin C	Vitamin C enhances non-haem iron absorption	Combine plant-source iron with vit C
Magnesium + Antibiotics	Magnesium may reduce absorption of certain antibiotics (tetracyclines, quinolones)	Separate by 2 hours
B12 + Metformin	Metformin reduces B12 absorption over long term	Monitor B12 levels; supplement if needed
K2 + Warfarin	K2 directly antagonises warfarin anticoagulation	Contraindicated — consult physician

Combination	Interaction	Recommendation
Ashwagandha + Thyroid meds	May increase thyroid hormone levels	Monitor thyroid function; inform doctor

3.7 Supplement Dosage Quick Reference

Supplement	Daily Dose	Timing	Key Benefit	Main Risk
Creatine Monohydrate	3–5 g	Anytime (post-workout ideal)	Strength, power, muscle mass	Water retention; GI if large dose
Whey Protein	20–40 g/serving	Post-workout / between meals	Protein, muscle recovery	Lactose intolerance; acne
Multivitamin	Per label	With breakfast	Micronutrient insurance	Fat-soluble vitamin toxicity
Pre-Workout (caffeine)	3–6 mg caffeine/kg BW	30–45 min pre-training	Focus, power, endurance	Insomnia, anxiety, dependence
BCAAs	5–10 g	During/post fasted training	Reduce catabolism	Redundant if protein adequate
Beta-Alanine	3.2–6.4 g	Daily (split doses)	60–240 sec endurance	Paraesthesia (harmless)
Fish Oil (Omega-3)	1–4 g EPA+DHA	With meals	Inflammation, heart, brain	Fishy breath; bleeding risk
Vitamin D3	1000–4000 IU	With fatty meal	Bone, immunity, testosterone	Hypercalcaemia at mega-doses
Vitamin K2 (MK-7)	100–200 mcg	With fatty meal	Bone calcium direction	Warfarin interaction
Magnesium Glycinate	200–400 mg	Evening	Sleep, cramps, recovery	Diarrhoea (high doses)
Ashwagandha	300–600 mg	Anytime with food	Cortisol, stress, recovery	GI; autoimmune caution
L-Glutamine	5–10 g	Post-workout / pre-bed	Gut health, immunity	Generally very safe
Zinc	8–25 mg	Evening with food	Immunity, testosterone	Nausea; copper depletion
Probiotics	≥10 billion CFU	With food	Gut health, immunity	Initial bloating
L-Theanine	100–200 mg	Pre-focus tasks / pre-sleep	Calm focus, anxiety	Very safe; mild drowsiness

IMPORTANT DISCLAIMER

This document is intended for educational and RAG reference purposes only. It does not constitute medical advice. Individual responses to supplements vary. Always consult a qualified healthcare professional before beginning any supplement programme, especially if you have underlying health conditions, are pregnant or breastfeeding, or are taking prescription medications. Supplement claims are subject to regulatory variation by country.